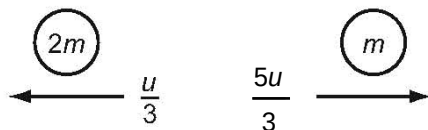


- 6 The diagram shows two spherical masses approaching each other head-on at the same speed u . One has mass $2m$ and the other has mass m . The masses then collide elastically.

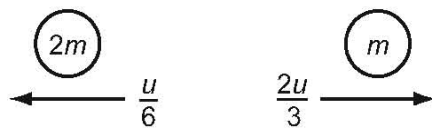


Which diagram shows the speeds of the masses after the collision?

A



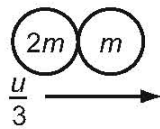
B



C



D



Ans: A

Since collision is elastic,

$$u - (-u) = v_2 - v_1$$

$$2u = v_2 - v_1 \quad (\text{eliminates B, C \& D})$$

OR

By conservation of momentum,

$$2mu - mu = 2mv_1 + mv_2$$

$$u = 2v_1 + v_2 \quad (\text{eliminates B, C \& D})$$

- 7 A beam, the weight of which may be neglected, is supported by three identical springs. When